

CHAPTER 10

Transaction Exposure

Managing Foreign Currency Risk in Global Business

Learning Objectives

10.1

Examine the three major types of foreign exchange exposure

10.2

Explore why firms choose to hedge foreign exchange exposure

10.3

Define and measure transaction exposure

10.4

Describe how a company hedges its transaction exposures (Ganado case)

10.5

Detail how FX risk management is conducted by firms in practice

SECTION 1

Types of FX Exposure

Understanding what changes when exchange rates move

What Is Foreign Exchange Exposure?

- The potential for a firm's profitability, net cash flow, and market value to change due to exchange rate movements.
- Two broad categories: (1) Accounting exposures — transaction & translation | (2) Economic exposure — operating
- Key task of the financial manager: MEASURE exposure and MANAGE it to maximize firm value.
- Currency risk = variance in expected cash flows caused by unexpected exchange rate changes.

Transaction vs. Operating Exposure



Transaction Exposure

- Deals with ALREADY CONTRACTED cash flows
- Arises from existing A/R, A/P in foreign currency
- Short-term; settlement date known
- FX rate changes → direct gain or loss
- Example: US firm billed £1M — dollar value uncertain until collected



Operating Exposure

- Deals with EXPECTED (not yet contracted) cash flows
- Arises from FX effects on competitiveness & pricing
- Long-term, strategic in nature
- Example: EUR weakens → German competitor undercuts US exporter on price
- Also called: economic, competitive, or strategic exposure

Translation Exposure – Key Facts



Definition

Potential accounting change in shareholders' equity when translating foreign subsidiaries' statements into the parent's reporting currency for consolidation.



Cash Flow Impact



NONE — translation exposure is purely an accounting change. It does NOT change cash flows or create tax liabilities.



When It Occurs



Each time the firm prepares consolidated financial statements and the foreign subsidiary's currency has moved against the reporting currency.



Common Confusion

Often confused with transaction exposure.
Remember: translation = accounting only.
Transaction = real cash flows already on the books.

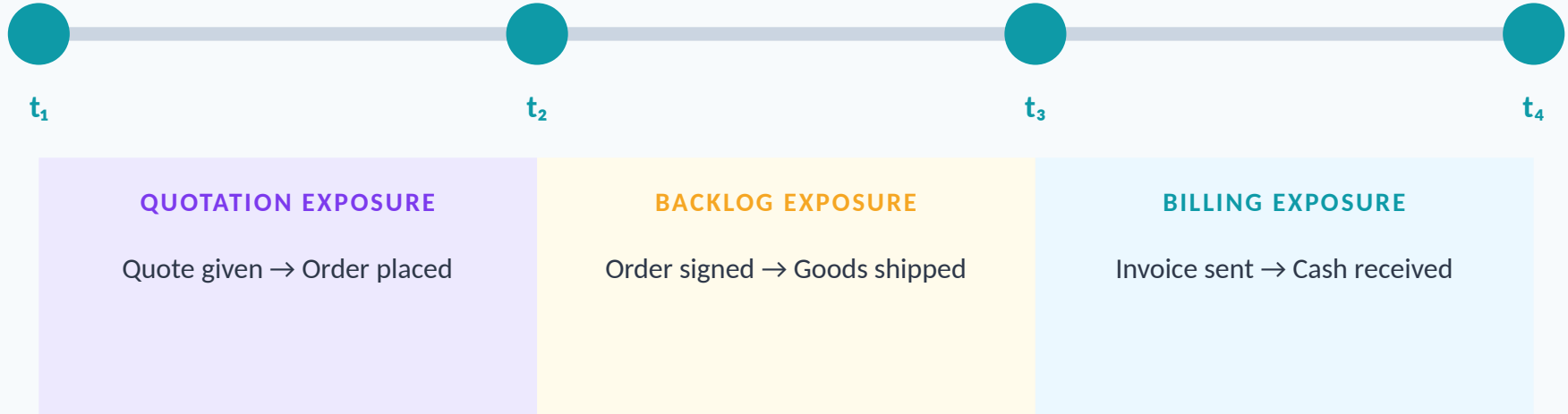
Life Span of Transaction Exposure

Seller quotes
price to buyer

Buyer places
firm order

Seller ships &
bills (A/R created)

Buyer pays
in cash



Most firms hedge only billing exposure. Some also hedge backlog. Very few hedge quotation exposure.

Transaction Exposure — GBP Receivable



SCENARIO

US Exporter (Ganado) quotes a price to a UK buyer on Jan 1 (t_1). UK buyer confirms order Jan 15 (t_2). Goods shipped and invoiced Feb 1 (t_3) — A/R of £1,000,000 created. Payment due Apr 1 (t_4).

Spot rate on Feb 1: \$1.7640/£ | Spot rate on Apr 1 (actual): \$1.70/£



RESULT & TAKEAWAY

USD received = £1,000,000 × \$1.70 = \$1,700,000

Expected at booking (Feb 1): £1,000,000 × \$1.7640 = \$1,764,000

Transaction LOSS = \$64,000 due to GBP depreciation over 59 days.

→ This is **BILLING EXPOSURE**: the loss occurred between invoicing (t_3) and payment (t_4).

→ A forward hedge or option on Feb 1 could have locked in a rate near \$1.7540.

SECTION 2

Why Hedge?

Arguments for and against managing currency risk

What Is Hedging?



Definition



Taking a position (asset, cash flow, or contract) that **RISES** in value when your existing exposure **FALLS** — and vice versa. The goal is to offset potential losses from adverse FX moves.



What It Reduces



Hedging reduces the **VARIANCE** of expected cash flows around the mean. It narrows the distribution — reducing both the downside and the upside.



What It Does NOT Do



Hedging does **NOT** increase expected cash flows. The average (mean) outcome is unchanged. It only reduces how much the actual result can deviate from that average.



The Trade-Off

By eliminating downside risk, hedging also eliminates upside gains from favorable rate moves. This is the core trade-off every treasurer must understand.

Should Firms Hedge? The Debate

✓ Arguments FOR Hedging

- Improves cash flow predictability → better planning
- Reduces probability of financial distress (cash flow below minimum)
- Management knows the firm's FX risk better than shareholders
- Management can exploit FX market disequilibrium

✗ Arguments AGAINST Hedging

- Shareholders can diversify FX risk themselves via portfolio
- Does NOT increase expected cash flows — only reduces variance
- Managers may hedge for accounting appearances, not value
- Managers cannot consistently outguess the market
- Adds cost; from efficient market view, may add no value

Why Hedging Creates Value — Cash Flow Distribution



SCENARIO

A US multinational expects net cash flows of \$5M next year. Unhedged, actual results range \$2M–\$8M depending on FX rates. The firm needs at least \$3.5M to service debt (the financial distress threshold).



RESULT & TAKEAWAY

UNHEDGED: 20% probability cash flows fall below \$3.5M → risk of default/distress.

HEDGED (forward contracts): range narrows to \$4M–\$6M. Probability of falling below \$3.5M drops near zero.

→ Expected value stays \$5M but VARIANCE shrinks dramatically.

→ Lower distress risk can reduce the firm's cost of capital — creating real economic value through hedging.

SECTION 3

Ganado Corporation Case

Four hedging alternatives for a £1,000,000 receivable

The Setup: Ganado's £1,000,000 A/R

Spot Rate

\$1.7640 / £

90-day Forward

\$1.7540 / £

US Borrow Rate

8% p.a. = 2%/90d

US Invest Rate

6% p.a. = 1.5%/90d

UK Borrow Rate

10% p.a. = 2.5%/90d

UK Invest Rate

8% p.a. = 2%/90d

Put Option: Strike = \$1.75/£ | Premium = 1.5% of notional | Maturity = 90 days

CFO Maria Gonzalez's 4 Alternatives:

1

Remain Unhedged

2

Forward Market

3

Money Market

4

Options Market

Alternative 2: Forward Market Hedge

1

Sell Forward: Enter a 90-day forward contract to SELL £1,000,000 at the locked-in rate of \$1.7540/£



2

Collect A/R: In 90 days, receive £1,000,000 from the UK buyer as payment



3

Deliver & Receive: Deliver the £1,000,000 to the bank under the forward contract



4

USD Receipt: Receive \$1,754,000 ($= £1,000,000 \times \1.7540) — guaranteed, regardless of spot rate



Locked-in receipt: \$1,754,000 | No FX risk remains | Upside foregone if £ strengthens

Alternative 1: Remain Unhedged



SCENARIO

Ganado takes NO hedging action. It collects £1,000,000 in 90 days and converts at whatever spot rate prevails on that date.

Forecast spot (in 90 days) = \$1.7600/£ → Expected receipts = \$1,760,000



RESULT & TAKEAWAY

If £ STRENGTHENS to \$1.85 → Ganado receives \$1,850,000 (UPSIDE: +\$90,000 vs forecast)

If £ WEAKENS to \$1.65 → Ganado receives \$1,650,000 (LOSS: -\$110,000 vs forecast)

→ Full exposure in both directions. Choose ONLY if management has a strong view that £ will strengthen AND the firm has high risk tolerance.

Alternative 3: Money Market Hedge

1

Borrow £ today: Borrow £975,610 at UK rate of 10% p.a. (2.5%/90d) → In 90 days repay exactly $£975,610 \times 1.025 = £1,000,000$

↓

2

Convert to USD: Sell £975,610 at today's spot (\$1.7640) → receive \$1,720,976 immediately

↓

3

Invest USD: Invest \$1,720,976 at US rate of 6% p.a. (1.5%/90d) for 90 days

↓

4

Maturity: USD investment grows to $\$1,720,976 \times 1.015 = \$1,747,791$ | Repay GBP loan using A/R collected

✓ Effective receipt: ~\$1,747,791 | Compare to forward hedge \$1,754,000 → forward hedge is better here

Alternative 4: Options Market Hedge (Put Option)

BUY a put option: right to sell £1,000,000 at \$1.75/£ | Premium = $1.5\% \times £1\text{M} \times \$1.7640 = \$26,460$ | Maturity = 90 days

Scenario: £ weakens → \$1.65/£

Action: EXERCISE the put (strike \$1.75 > spot \$1.65) → Net: $£1\text{M} \times \$1.75 - \$26,460 = \$1,723,540$

Scenario: £ stable → \$1.75/£

Action: Indifferent — exercise or transact at spot → Net: $£1\text{M} \times \$1.75 - \$26,460 = \$1,723,540$

Scenario: £ strengthens → \$1.85/£

Action: LET PUT EXPIRE — sell £ at spot instead → Net: $£1\text{M} \times \$1.85 - \$26,460 = \$1,823,540$ 🎉 upside!

✓ Key Advantage: Options provide a FLOOR (downside protection) while keeping upside open. Cost = premium.

Hedging Alternatives: Summary Comparison (A/R)

Strategy	Instrument	Certainty of \$ Amount	Upside Kept?	Cost
Unhedged	None	None – full FX risk	Yes	None
Forward Hedge	Forward contract	High – rate locked in	No	Opportunity cost
Money Market Hedge	Loan + deposit	High – effectively locked	No	Interest rate spread
Options Hedge (Put)	Put option	Floor guaranteed; upside open	Yes	Option premium

Hedging a Payable (A/P) — Mirror of A/R

Ganado owes £1,000,000 to a UK supplier in 90 days. RISK: if £ strengthens vs \$, paying in USD costs MORE.

A/R (Receivable)
Fear: £ WEAKENS

Forward:

SELL £ forward

Option:

BUY a PUT on £
(right to sell £)

A/P (Payable)
Fear: £ STRENGTHENS

Forward:

BUY £ forward

Option:

BUY a CALL on £
(right to buy £)

Memory rule: RECEIVABLE → SELL hedge (put option or forward sell). PAYABLE → BUY hedge (call option or forward buy).

SECTION 4

Natural & Financial Hedges

Hedging without derivatives

Natural vs. Financial Hedges

Natural (Operating) Hedge

- Arises from normal business operations — no derivative needed
- A/R and A/P in same currency offset each other
- Revenues and costs denominated in same foreign currency
- Example: US firm earns £1M (A/R) AND owes £1M (A/P) → net exposure = £0
- Usually the most cost-effective approach

Financial Hedge

- An offsetting financial instrument or obligation
- Types: forward, futures, options, swaps, foreign-currency debt
- Example: borrow in GBP to offset a GBP receivable
- The loan repayment is a liability that offsets the asset
- Care: distinguish operating hedges (operating CFs) from financial hedges (financing CFs)

Contractual Hedging Instruments



Forward Contract

OTC agreement to buy/sell FX at a set rate on a future date. Most common instrument. Eliminates FX risk completely but gives up upside. Highly customizable.



Futures Contract

Exchange-traded standardized forward. Marked to market daily. Less flexible but transparent pricing. Used mainly by financial institutions and large MNEs.



Options (Put/Call)

Right but NOT obligation to transact at the strike rate. Costs a premium but preserves upside. Most flexible hedging instrument for managing downside risk.



Money Market Hedge

Borrow in foreign currency + convert + invest in home currency. Replicates a forward hedge using loans and deposits. Useful when forward markets are illiquid.

SECTION 5

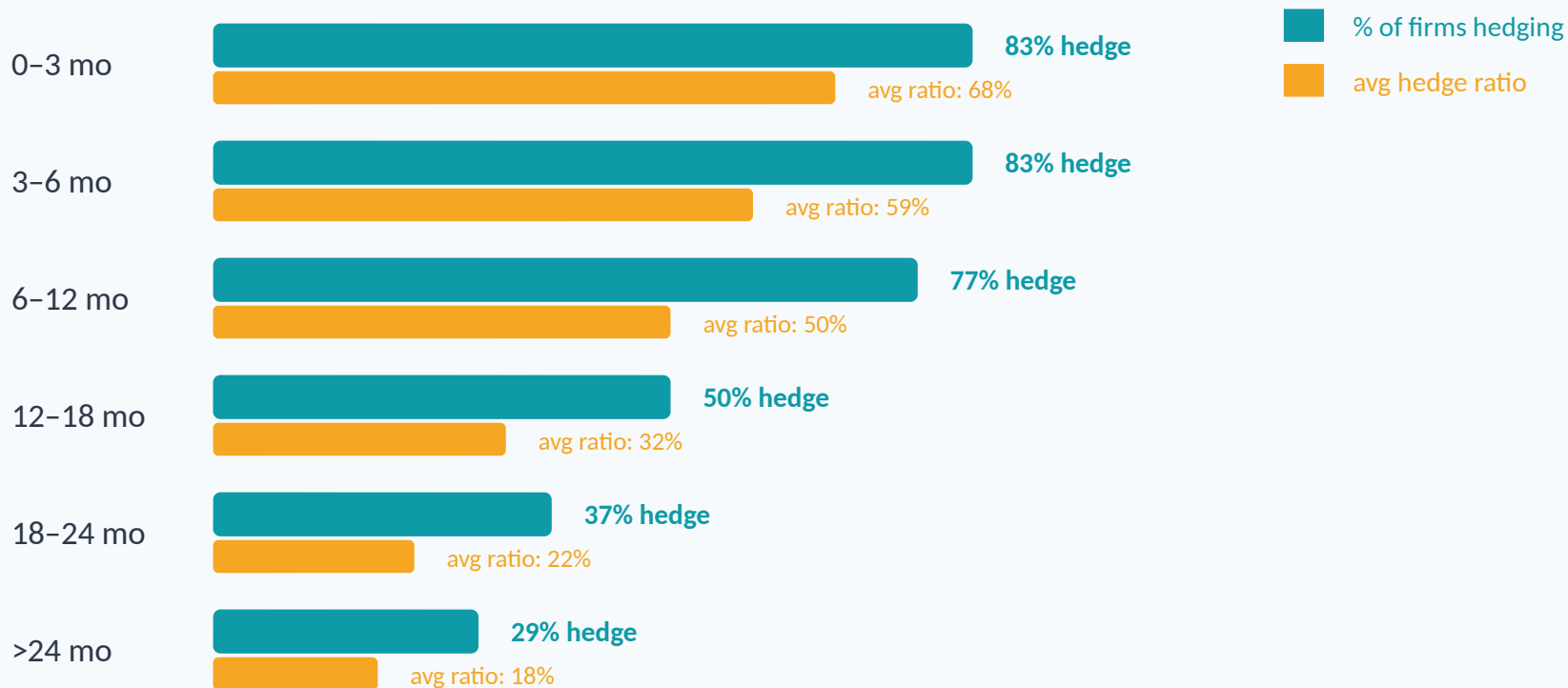
FX Risk Management in Practice

How real firms manage transaction exposure

The Treasury Function

- Treasury is typically a COST CENTER — not expected to add profit by speculating on FX rates.
- Currency risk managers are expected to be CONSERVATIVE — err on the cautious side.
- Firms must decide: Which exposures to hedge? (Quotation? Backlog? Billing? Anticipated?)
- Policy: Many firms do NOT hedge quotation or backlog exposure — only transactions on the accounting books.
- Trend: An increasing number of MNEs actively hedge backlog and even anticipated exposures selectively.
- Anticipated exposures = transactions with no formal contract yet between parties (e.g., forecast sales).

Survey: How Firms Hedge by Maturity (Deloitte 2016)



KEY INSIGHT: Near-term exposures are hedged most aggressively. Certainty about exposure decreases with longer horizons → firms hedge less.

Forward Contracts — How Firms Use Them

- Forward contracts remain the PRIMARY instrument for transaction exposure management worldwide.
- Rule 1: When forward rate locks in a GAIN vs booked rate → 100% coverage is common policy.
- Rule 2: When forward premium/discount is small (1–2%) → high forward cover is still required.
- Rule 3: Firms that don't use options rely EXCLUSIVELY on forwards + money market hedges.
- Why firms AVOID options: (a) Cost of premium, (b) Complexity, (c) Options only bound the exposure — they don't eliminate it.
- Proportional hedging: MNEs mandate set hedge % per currency and maturity bucket (e.g., 100% for 0–3 mo, 75% for 3–6 mo).

SECTION 6

Appendix: Complex Option Hedges

Advanced strategies for receivable exposure management

Four Complex Option Strategies



1. Simple Put Hedge



Buy a put option on £1M. Sets a floor on dollar receipts. Two strike choices (\$1.44 vs \$1.47) offer different premium/protection trade-offs. Higher strike = better floor but costs more premium.



2. Synthetic Forward



SELL a put + BUY a call at the SAME strike. Net premium $\approx$ zero. Replicates a forward contract — locks in both floor AND ceiling at the strike price. No upside.



3. Range Forward (Collar)

Buy put at lower strike (\$1.45) + sell call at higher strike (\$1.49) + Buy a currency. Protects downside, caps upside. Net premium is low. Outcome is bounded between the two strikes.



4. Participating Forward



Buy put + sell call on a SMALLER notional (531,765 vs 1M). Full downside floor. Participation rate $\approx$ 46.82% of any upside above strike. Low cost, partial flexibility.

Complex Option Hedges – Comparison

Strategy	Construction	Downside Protection	Upside Participation	Net Premium Cost
Simple Put	Buy put option	Full floor at strike	100% above strike	Option premium paid
Synthetic Forward	Buy put + Sell call (same strike)	Full floor at strike	None (capped)	Near zero
Range Forward	Buy put + Sell call (diff strikes)	Floor at put strike	Partial (up to call strike)	Low (premium offset)
Participating Forward	Buy put + Sell call (smaller notional)	Full floor at put strike	~47% of upside above strike	Low

SECTION 

Practice Questions

Test your understanding of Chapter 10



Which type of FX exposure creates only accounting changes with NO impact on cash flows or taxes?

A

Transaction exposure

B

Operating exposure

C

Translation (accounting) exposure

D

Quotation exposure

Translation exposure only affects the consolidated balance sheet. No cash changes hands.



The DIFFERENCE between transaction exposure and operating exposure is that transaction exposure deals with cash flows that are:

A

Expected but not yet contracted

B

Already contracted (existing obligations)

C

Long-term and strategic in nature

D

Related to translating foreign subsidiary statements



Transaction = contracted. Operating = expected future CFs not yet contracted.

 **Ganado has a £1,000,000 A/R. It enters a 90-day forward contract to sell £1M at \$1.7540/£. What is the guaranteed USD receipt?**

A \$1,764,000 (spot rate $\times$ notional)

B \$1,760,000 (forecast spot)

C \$1,754,000 (forward rate $\times$ notional)

D Cannot be determined

 $\$1,000,000 \times \$1.7540 = \$1,754,000$. The forward rate is locked in — no FX risk remains.

In the money market hedge, Ganado borrows £975,610 at 2.5%/90d and converts at \$1.7640. It invests at 1.5%/90d. Approximate USD received after 90 days?

A

\$1,720,976

B

\$1,747,791

C

\$1,754,000

D

\$1,764,000

💡 $£975,610 \times 1.7640 = \$1,720,976 \rightarrow \text{invested at } 1.5\% \rightarrow \$1,720,976 \times 1.015 \approx \$1,747,791.$

 **Ganado buys a £1M put option at \$1.75/£ (premium \$26,460). The spot rate in 90 days is \$1.85/£. What should Ganado do?**

A Exercise the put; sell at \$1.75/£

B Let the put expire; sell £1M at spot \$1.85/£

C Buy a call option immediately

D Enter a forward contract now

 Spot (\$1.85) > strike (\$1.75) → let put expire; sell at spot. Net = \$1,850,000 - \$26,460 = \$1,823,540.



A firm has a ¥50,000,000 PAYABLE in 90 days. Which option should it buy to hedge?

A

A put option on yen (right to SELL ¥)

B

A call option on yen (right to BUY ¥)

C

A put option on USD

D

No option needed — payables are naturally hedged



A/P → firm must BUY yen. A CALL gives the right to buy ¥ at the strike — caps the cost if ¥ strengthens.



Which is the strongest argument AGAINST corporate hedging?

A

Hedging reduces variance of cash flows

B

Managers have better knowledge of FX risk

C

Shareholders can diversify FX risk themselves in their own portfolios

D

Hedging reduces likelihood of financial distress



Core finance theory: if investors can diversify currency risk, corporate hedging adds no value and only costs money.



A Synthetic Forward is constructed by:

A

Buying a put AND a call at different strikes

B

Buying a put AND selling a call at the **SAME** strike

C

Selling a put AND selling a call

D

Buying two put options at the same strike



Synthetic Forward = Long Put + Short Call at same strike. Net premium ≈ 0 . Locks in floor = ceiling = strike.

Chapter 10 — Key Takeaways

→ FX exposure = 3 types: Transaction (contracted CFs), Operating (expected CFs), Translation (accounting only).

→ Hedging reduces VARIANCE of cash flows — it does NOT increase expected returns.

→ Transaction exposure spans three stages: Quotation → Backlog → Billing.

→ 4 strategies for A/R: Unhedged | Forward Hedge | Money Market Hedge | Options Hedge.

→ Options are the ONLY hedge that preserves upside — at the cost of a premium.

→ In practice: forwards dominate; firms hedge near-term exposures most aggressively.